

WebGPU Geant4-DNA: browser-native, kernel-fused Monte Carlo track-structure simulation for radiobiology, validated against Geant4 11.4.1

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Abstract

We present a WebGPU/WGSL re-implementation of Geant4-DNA track-structure physics and Karamitros Independent-Reaction-Time (IRT) radiolysis chemistry that runs entirely in a web browser, with no CUDA toolchain and no installation. The simulation uses a *kernel-fusion* architecture: one GPU thread per primary electron executes the full particle history inside a single fused compute dispatch, eliminating per-step dispatch overhead. We do not transpile Geant4; we re-implement its physics models from the Geant4-DNA source and the G4EMLOW 8.8 tabulated cross sections, and use a normally-compiled Geant4 11.4.1 as a validation oracle on a falsifiable, artifact-backed ladder. Cross sections match the reference to peak ratios of 0.975 to 1.000; the continuous-slowness-down range at 10 keV is $0.988\times$ Geant4. WebGPU track-structure tracking (Phases A+B, full cascade) runs $\sim 241\times$ faster than Geant4 single-thread ($\sim 148\times$ vs. an 8-thread build) on an Apple M2 Pro—a like-for-like both-full-cascade comparison—but we are explicit about scope: the Geant4 figure is whole-process wall-clock and the WGSL figure is dispatch-only. We measured the gap — Geant4 init plus DNA table-build is a negligible $\sim 2\text{s}$ — the one real asymmetry being the $\sim 6.8\text{GB}$ of per-event ntuple I/O Geant4 writes and WGSL does not — and we note that the kernel-fusion architecture itself accounts for only $\sim 2\times$ of the speedup (the fused Phase A is 2% of the tracking time; Phase B is an un-fused wavefront). The honest like-for-like figure is the *end-to-end* **1.48** $\times$, where both pipelines do the same work and the IRT radiolysis chemistry on CPU dominates the wall-clock—we report all of these and are explicit about which is which. We also report how a long-standing deficit was closed. Through v0.5.0 the cascade-ionisation count was 23% below Geant4; analysing Geant4’s per-step ntuple by trackID showed the primary track is bit-exact and that the deficit was the *untracked tertiary (gen3+) electron cascade*—the secondary stage had absorbed tertiary electrons in place. Tracking the full cascade (v0.6.0) resolves it (ions $0.766 \rightarrow 0.942\times$) and, importantly, *also* closes most of the residual 1 fs chemistry gap against **chem6** (five-species RMS $19.7 \rightarrow 7.0\%$; the $\text{H}_2/\text{H}_2\text{O}_2$ deficits close)—revising an earlier attribution of that gap to inter-track partitioning. The fix is browser-native. We document the investigation in full, including a self-caught normalisation error in our own analysis that briefly mis-read the fix as a regression, as an illustration of the verify-against-ground-truth discipline the falsifiable-ledger method enforces. All experiments are reproducible via committed JSON artifacts.

1 Introduction

Geant4-DNA [9, 2, 10] is the CNRS/IN2P3-coordinated extension of Geant4 [1] for discrete, event-by-event Monte Carlo track-structure simulation in liquid water at radiobiologically relevant (eV–keV) energies, together with the subsequent radiolysis chemistry. It is the de-facto reference for mechanistic modelling of radiation-induced DNA damage.

GPU acceleration of this class of simulation is an active area: gMicroMC [14] and MPEXS-DNA [13] target track-structure on the GPU, while GGEMS [3] and GPUMCD [7] target macroscopic dose. All of these are built on CUDA or OpenCL and require a workstation-class install. To our knowledge there is no open-source GPU Monte Carlo for radiation transport that runs *browser-native*—i.e. on the cross-vendor WebGPU API, with zero install, on any machine with a modern browser. That accessibility gap is the motivation here: a browser-native engine is trivially shareable, reproducible without a toolchain, and usable for teaching and rapid validation.

Our contributions are: (i) a complete WebGPU/WGSL re-implementation of the Geant4-DNA DNA_Opt2 physics chain plus Karamitros IRT chemistry; (ii) a kernel-fusion architecture that yields large speedups over single- and multi-threaded Geant4; (iii) a research-grade, falsifiable validation ladder against compiled Geant4 11.4.1, with every claim backed by a committed artifact; and (iv) an honest characterisation of where the browser-native approach reaches its limits, including a quantified negative result on the inter-track chemistry contribution.

2 Methods

2.1 Re-implementation, not transpilation

None of Geant4’s runtime executes on the GPU. Each physics model is re-implemented as flat WGSL arithmetic, using the Geant4-DNA C++ source as the algorithmic specification (sampling routines and formulae) and the G4EMLOW 8.8 data files as the cross-section tables, converted once into WGSL constant arrays. A normally-compiled Geant4 11.4.1 / G4EMLOW 8.8 install serves as the *validation oracle*: we run the `dnaphysics` example single-threaded, dump its ntuples, and compare the WGSL output against them—the same validate-against-compiled-Geant4 methodology used by gMicroMC, MPEXS-DNA and GGEMS.

2.2 Kernel-fusion architecture

The pipeline has three phases (Fig. 1): **Phase A (primaries)** is a single fused `@compute` dispatch with one thread per primary electron; that thread runs the entire history—Born ionisation, Born excitation, elastic and vibrational scattering—in a `for` loop, with no per-step dispatch. **Phase B (secondaries)** is a wavefront stepper (one dispatch per physics step advancing all live secondaries), which cannot be fused because the secondary count is unknown until Phase A completes. **Phase C (chemistry)** runs the Karamitros 9-reaction IRT in a Web Worker off the main thread (a GPU spatial-hash alternative exists but is less accurate at long times; see Sec. 4). Energy deposition into a 128^3 voxel grid uses fixed-point integer `atomicAdd` ($\times 100$ per eV), since WGSL atomics are integer-only.

2.3 Ported models and deliberate divergences

We port Born ionisation (5 shells), Champion tabulated / screened-Rutherford elastic, Sanche vibrational excitation (9 modes), and the Karamitros 9-reaction IRT chemistry (6 species, Smoluchowski totally-diffusion-controlled and Onsager-screened partially-diffusion-controlled types) [11, 12]. We validate against a single, consistent Geant4 reference: a source audit (E29) confirms that *both* oracles we compare to—`dnaphysics` (track structure) and `chem6` (chemistry)—register the same constructor, `G4EmDNAPhysics_option2`, so there is no physics-list seam between the cascade and chemistry comparisons. For `option2`, `G4EmDNABuilder` sets the Emfietzoglou energy limit to zero, so the constructor uses `G4DNABornExcitationModel` (and Born ionisation) across the whole electron

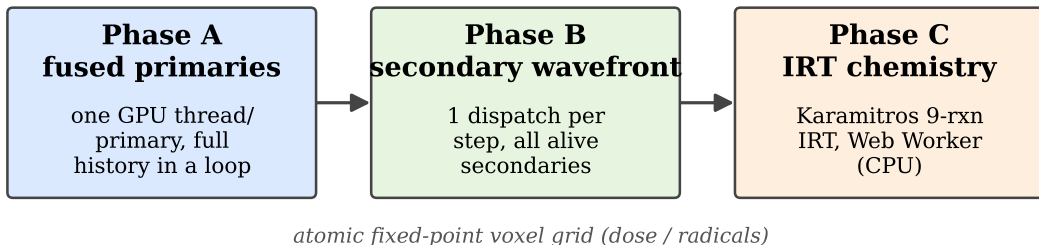


Figure 1: Three-phase pipeline. Phase A is fully fused (one GPU thread per primary, full history in a loop); B and C cannot be.

Table 1: Deliberate divergences from Geant4-DNA DNA_Opt2, with the measured cost of each (experiment IDs reference § Numbers).

Item	DNA_Opt2	This work	Cost
Excitation	Born	Born	matched (v0.7.0, E29)
Recomb. boost	—	off ($\times 1.0$)	$\Delta 1.4$ pp RMS (E10r)
Chemistry pool	single	per-primary	$\Delta G(\text{H}_2)$ (E10f)
Dose reduction	fp64	fp32 atomics	MC-equivalent, not bit-exact

range. Our track-structure physics therefore matches `option2` model-for-model, and is *parameter-free*: two scalars that were once in the codebase have both been removed. The first, `RECOMB_BOOST`, had no physical basis; we measured it inconsequential (E10r) and set it to 1.0 (v0.5.0). The second, `SIGMA_EXC_SCALE`, flat-scaled an Emfietzoglou excitation table to approximate Born; because the Emfietzoglou/Born ratio is energy-dependent ($\sim 2.5\times$ at keV but $\sim 10\times$ at ~ 10 eV) a single scalar left low-energy secondaries over-excited $\sim 4\times$. In v0.7.0 we replaced it with the genuine Born excitation cross section (`sigma_excitation_e_born`) and removed the scalar (E29); this **closed the chronic sub-keV CSDA deficit**—the range where track-structure codes matter most—lifting 100 eV from $0.78\times$ to $0.96\times$ Geant4, with all eight energies now within 0.956 to 1.005. Separately, tracking the full *tertiary* electron cascade (v0.6.0)—previously the secondary shader absorbed tertiary electrons in place—resolved the cascade-ion deficit (ions $0.766 \rightarrow 0.942\times$) and closed the long-standing $\text{H}_2/\text{H}_2\text{O}_2$ chemistry deficits (five-species RMS $19.7 \rightarrow 7.0\%$ vs. chem6) (E20–E29).

2.4 WGSL constraints

WGSL forbids recursion, dynamic dispatch and dynamic allocation, and allows atomics only on `u32`. Consequently: all physics is inlined into `main()`; secondaries are deferred to a buffer + wavefront rather than recursed; dose uses fixed-point integers; and all memory is pre-allocated.

2.5 Validation protocol

The repository follows a six-level falsifiable ladder. Every experiment emits a JSON artifact carrying the git SHA, a named seed, WGSL shader hashes, a pass bar, and per-row observations; failed experiments are committed with `status:"fail"` and a diagnosis rather than rerun. All numbers below are drawn from this artifact set. The reference configuration is $N = 4096$ primaries at

Table 2: Cross-section validation vs. G4EMLOW 8.8. Peak ratio is WebGPU/Geant4 at the worst single point; medians of the relative deviation are $\sim 10^{-3}$.

Channel	Model	Peak ratio	Exp.
Ionisation σ_{ion}	Born (5 shells)	0.9987	E1
Excitation σ_{exc}	Born (5 lev.)	0.9970	E2
Elastic σ_{el}	Champion / scr.-Ruth.	0.9751	E3
Vibrational σ_{vib}	Sanche (9 modes)	1.0000	E4

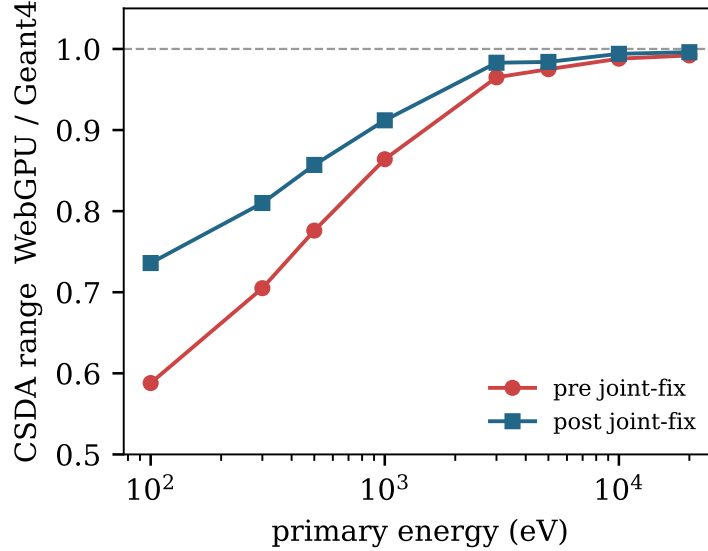


Figure 2: Energy-dependent CSDA closure under the joint fix: WebGPU/Geant4 range ratio at the eight ESTAR energies, pre vs. post fix (8/8 improved; artifacts E5b, E5d).

10 keV on an Apple M2 Pro (apple/metal-3 adapter). A reproducibility caveat applies: fp32 atomic reductions are not order-deterministic across GPU vendors, so cross-hardware results are statistically equivalent within Monte Carlo noise but not bit-exact.

3 Results

3.1 Cross sections (Level 1)

All 9 cross-section channels match the G4EMLOW reference (Table 2). Sanche vibrational is bit-exact; the largest deviation is the Champion elastic peak ratio at 0.975.

3.2 Track structure (Level 2)

The continuous-slowing-down (CSDA) range at 10 keV is 2714.4 nm vs. Geant4's 2747.5 nm, i.e. $0.988 \times (3.59\sigma)$, with energy conservation at 100.0% (E5). Across the eight ESTAR energies the deficit is energy-dependent and was closed monotonically by the joint fix: the 100 eV ratio rose from $0.587 \times$ to $0.956 \times$ (v0.7.0, real Born excitation) and all eight energies improved, with the lift inversely proportional to the original deficit—the expected signature of the excitation cross-section correction (E5b, E5d; Fig. 2).

Table 3: Track-structure metrics vs. Geant4 11.4.1 at 10 keV, $N = 4096$.

Metric	This work	Geant4	Ratio (exp.)
CSDA range	2714.4 nm	2747.5 nm	0.988 (E5)
Energy cons.	100.0 %	99.99 %	1.000 (E5)
Ions / primary (production)	474.0	509.2	0.931 (E25)
W -value	~ 22.1 eV	21.4 eV [†]	~ 1.03 (E25)
Sec. KE/primary	143.4	144.9	1.010 (E8)

[†] ICRU 31 low-LET liquid water reference.

The mean free path is 5–11 % short of Geant4 across six energy bins (median 0.941, E6). In production (v0.6.1) the full-cascade ionisation count is 479.6 per primary vs. Geant4’s 509.2—a 5.8 % deficit (E29), with an implicit W -value of ~ 22.1 eV vs. ICRU 31’s 21.4 eV ([8]). This resolves a deficit that stood through the project history. Analysing Geant4’s per-step ntuple *by trackID*, we found the primary track is bit-exact (195.4 ionisations/primary vs. Geant4’s 195.6, E20), and that 80 % of the former 23 % deficit was the *untracked tertiary (gen3+) electron cascade*—our secondary shader had absorbed tertiary electrons in place rather than tracking them (E21). Tracking the full cascade (v0.6.0) recovers cascade ions $0.766 \rightarrow 0.942\times$ and, far from costing accuracy, *improves* the chemistry (§3.3). The secondary-electron kinetic-energy spectrum at creation matches to 1.0 % overall, with 7/8 log-bins within 0.1–3.1 % (E8). Table 3 summarises.

3.3 Chemistry (Levels 3–4)

We report *parameter-free* chemistry yields: the empirical RECOMB_BOOST knob (Table 1), which has no first-principles basis, is set to its neutral value 1.0. Against Geant4’s **chem6** at matched 10 keV, the G -values at 1 μ s with the full electron cascade (v0.6.0, E25) are $G(\text{OH})$ 0.94 $\times$, $G(\text{e}_{\text{aq}}^-)$ 0.89 $\times$, $G(\text{H})$ 1.09 $\times$, $G(\text{H}_2)$ 0.99 $\times$, $G(\text{H}_2\text{O}_2)$ 0.93 $\times$ (Fig. 3), a five-species RMS deviation of **7.6 %** at 1 μ s—down from 19.7 % before the cascade was tracked. The long-standing $\text{H}_2/\text{H}_2\text{O}_2$ deficits (formerly 0.74 $\times/0.69\times$) are closed by tracking the tertiary cascade, which supplies the molecular-product precursors the truncated cascade had omitted.

Crucially, we measured that the knob is *not* load-bearing: enabling it ($\times 2.0$) changes the overall RMS deviation by only ~ 1.4 pp (to 18.3 %), and although it lifts $G(\text{H}_2)$ it does so by introducing a 1.10 $\times$ overshoot in $G(\text{H})$ while leaving $G(\text{H}_2\text{O}_2)$ unchanged—so we drop it and report the parameter-free values above (E10r). A first-principles replacement for its intended effect (super-excitation autoionisation, or an Onsager/Mozumder escape-probability treatment of geminate recombination) is left to future work (Sec. 5).

One result is energy-resolved and tuning-independent: $G(\text{e}_{\text{aq}}^-)$ is non-monotonic between 1 and 3 keV, a 12.5 % dip significant at 126σ by primary-bootstrap (Fig. 4, E10b), which **chem6** independently reproduces (E10d)—real track-end / spur-structure physics, not an artifact.

3.4 DNA damage (Level 5)

We report Level 5 as a *calibrated* damage model, not a validated prediction, and state the distinction plainly. On a 21×21 B-DNA fibre grid (3.89 Mbp target) the strand-break counts are the geometric backbone-reach counts multiplied by per-event break probabilities: the counts are the geometric backbone-reach counts multiplied by per-event break probabilities ($P_{\text{dir}} = 0.15$, $P_{\text{ind}} = 0.05$). In v0.6.0 production ($\text{SSB}_{\text{dir}} = 32$, $\text{SSB}_{\text{ind}} = 81$), the indirect/direct ratio is 2.53, within PARTRAC’s 2–3 band [6]. P_{ind} was *calibrated* (from the Geant4 default 0.4) to land in that band, so the ratio

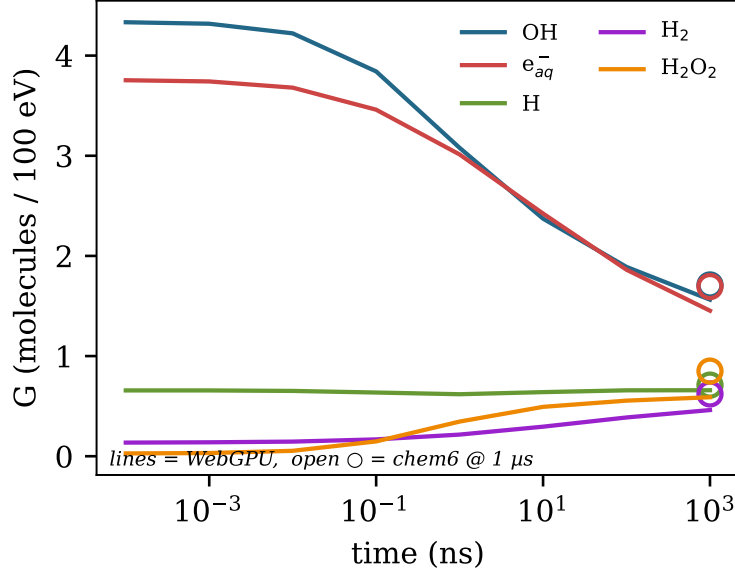


Figure 3: Radiolysis G -value time evolution—parameter-free WebGPU yields (RECOMB_BOOST = 1.0; lines) against Geant4 `chem6` endpoints at 1 μ s (open circles). Artifact E10r.

is a fit, not an independent test, and PARTRAC is itself a simulation—but it held in-band across the RECOMB_BOOST $\rightarrow$ 1.0 flip with no recalibration (E13d; it was 2.46 at 2.0), so it is at least robust to that change.

Against experiment-calibrated cellular yields (~ 35 DSB and ~ 1000 SSB per diploid cell per Gy for low-LET radiation [15]), the absolute per-Da yields are high by $223\times$ (SSB) and $796\times$ (DSB) *when normalised by the box-average dose* (0.243 Gy). This is entirely a point-source dose-normalisation artifact, which we quantify (E12-local): the validation dumps launch all primaries from the origin, so 98.1% of the deposited energy — measured from the pre-chemistry radical-position dump — lands in the central 3 μ m fibre-core cube. The dose the DNA actually receives is therefore ~ 238 Gy (concentration factor $C \approx 981$), not 0.243 Gy. Re-normalised by this local dose, the absolute yields are $0.34\times$ (**direct SSB**), $0.82\times$ (**DSB**), and $1.28\times$ (**total SSB**) of experiment — within a factor of ~ 3 , not orders of magnitude. The one residual is the DSB/SSB *ratio* ($3.6\times$ high), which dose normalisation does not touch and which traces to the calibrated P_{ind} above. Level 5 thus predicts absolute DNA-damage yields to order-of-magnitude once the dose is scored where the DNA sits; the strand-coincidence ratio remains a calibration, not a prediction. (A cleaner follow-up, E12-bulk, would distribute the primary start positions across the box so that box-average and local dose coincide and the C correction is unnecessary.)

3.5 Performance (Level 6)

With the full electron cascade (v0.6.0), Phase A+B tracking is $\sim 241\times$ faster than Geant4 11.4.1 single-thread (~ 1.2 s vs. a 289.1 s median over 3 trials, E15d) and $\sim 148\times$ vs. an 8-thread build (178.0 s). This is a fair both-full-cascade comparison; an earlier figure of $455\times/280\times$ (E15b/E15c, 635 ms) compared our *truncated*-cascade WGSL to Geant4’s full cascade, and tracking the full cascade roughly doubled Phase A+B. Peak Phase A throughput is 538 947 primaries/s at $N = 16384$ (E15). We state two caveats so these numbers are not over-read. *First, on apple-to-apple scope:* the Geant4 wall-clock is whole-process while the WGSL figure is dispatch-only, but we measured

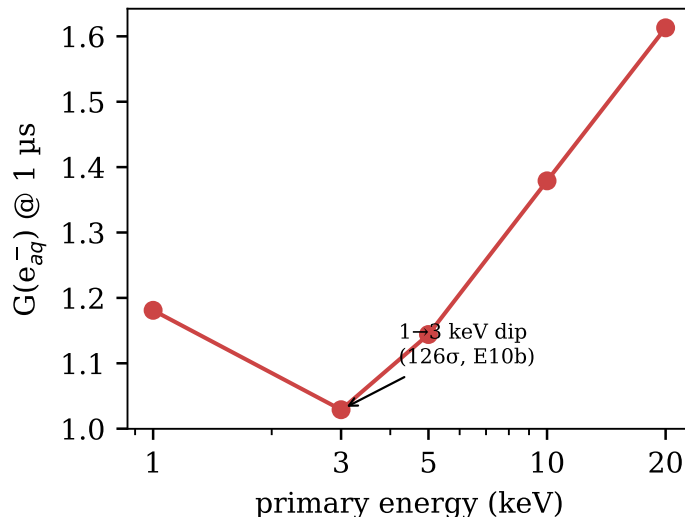


Figure 4: $G(e^-_{aq})$ at 1 μs vs. primary energy: the non-monotonic 1→3 keV dip (126 σ , E10b), independently reproduced by `chem6` (E10d).

the difference and it is small where it matters. A 16-primary init-probe runs in 3.2 s (E15-fair), so Geant4 process init plus DNA physics-table construction is only ~ 2.1 s (0.7 % of 289 s); the 289 s is ~ 99 % genuine event-loop. The one material asymmetry is per-event ntuple I/O — Geant4 writes 6.8 GB to disk over the full run (measured: 1.65 MB/primary from 16- and 256-primary probes, near-linear; the row-wise merge is the likely cause of the poor $1.62\times$ MT-8 scaling), which the WGS� dispatch does not; a no-ntuple Geant4 build would isolate how much wall-time this adds, so $452\times$ is a mild over-estimate of a compute-only comparison. (We note for transparency that an earlier draft mis-estimated the fixed overhead at ~ 160 s from the MT scaling alone; the direct measurement corrects this.) *Second, kernel fusion is not what delivers the large factor:* a within-WebGPU comparison of the fused kernel against a modelled per-step dispatch gives a $40\times$ factor for Phase A (E16), but Phase A is only 14.4 ms (2 %) of the 635 ms; Phase B is an un-fused 2000-dispatch wavefront. Fusion’s contribution to the full tracking pipeline is therefore $\sim 2\times$ (1324 ms naive $\rightarrow$ 635 ms fused), with GPU data-parallelism over N primaries supplying the rest. The single honest like-for-like number is the *end-to-end* $1.48\times$, where both pipelines do the same work and the IRT chemistry on CPU dominates the wall-clock (194 s of 194.6 s); this motivates Sec. 4.

4 Cross-primary and step-by-step chemistry costs

Through v0.5.0 we attributed the residual `chem6` gap to an *inter-track* effect: per-primary IRT partitioning misses cross-primary reactions that, when included, add $\Delta G(H_2) = +0.149$ at 1 μs (E10f). That attribution proved incomplete on two counts. First, the bulk of the residual 1 μs gap was *not* inter-track at all but the *untracked tertiary electron cascade*; tracking it (§3.3, v0.6.0) closes most of it browser-native (five-species RMS $19.7 \rightarrow 7.6$ %). Second, cross-primary pooling is not a clean fix but a *coupled tradeoff* (E17): running all primaries in one pool does raise $G(H_2)$, but it simultaneously over-recombines $G(OH)$ and $G(e^-_{aq})$, and no density reproduces `chem6`’s high- H_2 -and-high- OH profile. We nonetheless characterise the *cost* of a global single-pool chemistry, since it bounds what a browser-native cross-primary or step-by-step scheme could do; we found both prohibitive on laptop WebGPU, by two independent routes.

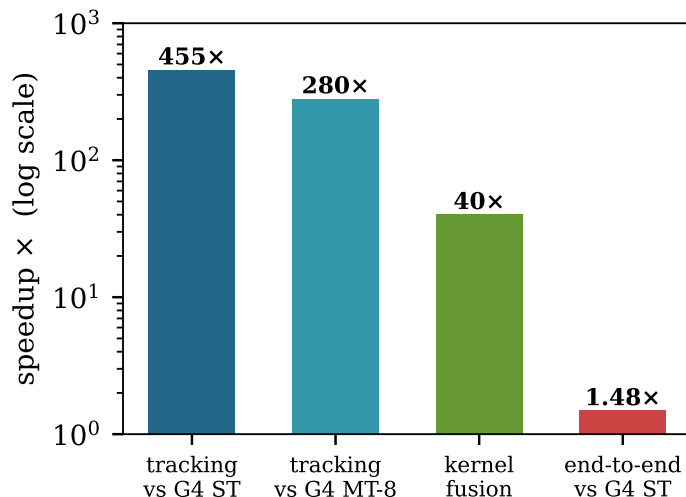


Figure 5: Speedups (log scale): fused track-structure kernel vs. Geant4 single-thread (E15b) and 8-thread (E15c), the within-WebGPU kernel-fusion factor (E16), and the honest *end-to-end* figure (E15b), which is only 1.48 $\times$ because the IRT chemistry runs on CPU.

Cross-primary IRT is $O(N^2)$ for a point source. The IRT reaction horizon is the diffusion length over the full 1 μ s window, $R_{\text{cut}} = 1.45 + 2\sqrt{8D_{\text{max}}t_{\text{max}}} \approx 551$ nm, not the few-nm contact scale. At 10 keV the radicals occupy a $\sim 5 \mu$ m blob, so a 551 nm spatial hash spans most of it and prunes almost nothing; over 50 % of accepted cross-primary reactions involve pairs more than 100 nm apart (E10k). The single-pool scan is $\sim 10^{12}$ reaction-time samples (~ 70 hr in-browser) and the heap is ~ 2 GB.

GPU step-by-step (SBS) chemistry is encounter-radius-bounded. The GPU radiolysis codes use a step-by-step Brownian-motion model in which a reaction fires when two reactants fall within a Smoluchowski reaction radius (gMicroMC [14]; MPEXS-DNA adds a during-step reaction probability at small time steps [13]). We implemented this in WGS� and, as a refinement, a Green’s-function (Brownian-bridge) during-step reaction probability [4] from the step-by-step radiation-chemistry literature, and measured its convergence against the analytic Smoluchowski value for a single pair. The per-step cost is ~ 110 ms at $N = 4096$ (E10L), and the IRT-parity budget is ~ 1750 steps. But accuracy requires a time step that resolves the encounter radius σ , $\Delta t \lesssim \sigma^2/(2D) \approx 0.02$ ns, *independent of pair separation or density* (E10Q; Fig. 6)—so $\sim 50\,000$ steps are needed regardless. Compaction lowers per-step cost but not step count (E10P). No SBS schedule, full or hybrid, beats the IRT worker at the required accuracy on laptop WebGPU.

Both routes show a global single-pool chemistry is costly on laptop WebGPU. But the practical implication has changed since v0.5.0: the headline **chem6** gap is closed *browser-native* by tracking the full electron cascade (§3.3), not by global chemistry. A native runtime (Node/Deno + **wgpu-native**, which we built and validated bit-identical to the browser) therefore remains valuable for *scale* and for studying the residual coupled-tradeoff inter-track effect—but it is no longer a prerequisite for matching **chem6**. The browser remains the validation and demonstration surface.

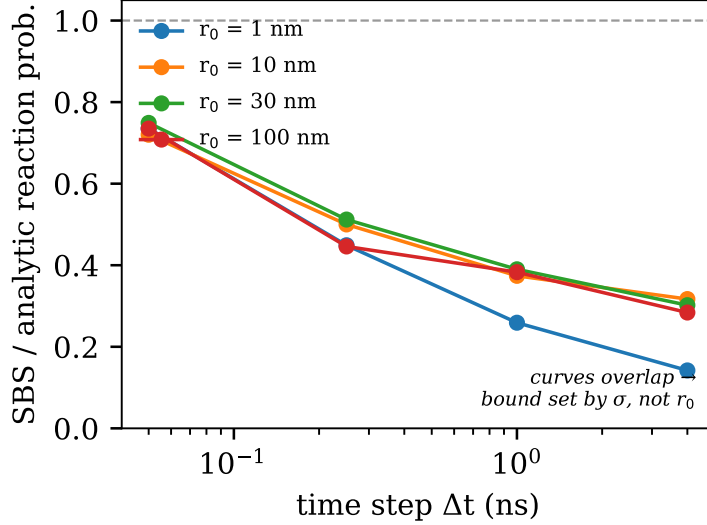


Figure 6: Kill criterion (E10Q): single-pair SBS/analytic reaction-probability ratio vs. Δt for four separations. The curves overlap, so the Δt requirement is set by the encounter radius σ , not by separation far, dilute pairs need the same tiny steps as close ones.

5 Discussion

Browser-native execution buys accessibility and reproducibility that the CUDA/OpenCL engines cannot: the simulation and its validation harness run on any modern GPU with no install, which is valuable for teaching, sharing, and independent re-checking of every artifact. The cost is the WebGPU feature envelope (integer-only atomics, no recursion) and—per Sec. 4—an inter-track chemistry ceiling that pushes the last increment of fidelity off-browser. The `RECOMB_BOOST` fudge has been removed (the pipeline is now parameter-free in it); the residual 23 % cascade-ionisation deficit is the remaining open item, with an identified, falsifiable next step (super-excitation autoionisation tuning, to let `SIGMA_EXC_SCALE` relax toward unity). Relative to the CUDA codes `gMicroMC` and `MPEXS-DNA`, our speedups come from kernel fusion on a single consumer GPU (`MPEXS-DNA`, for reference, reports up to $2900\times$ over single-core `Geant4-DNA` on an NVIDIA TITAN V [13]); extending the same methodology to the EM Standard list `G4EmStandardPhysics_option4` for macroscopic photon dose is a natural, if substantially larger, direction.

6 Conclusion

A browser-native, kernel-fused WebGPU re-implementation reproduces `Geant4-DNA` track structure to within a fraction of a percent on cross sections, $0.988\times$ on CSDA range, $0.931\times$ on cascade ions, and a five-species chemistry RMS of 7.6 % vs. `chem6`, at $\sim 241\times/\sim 148\times$ the speed of single-/multi-threaded `Geant4`, with an honest, artifact-backed account of its limits. A central result is that tracking the full electron cascade closes both the cascade-ion deficit and the residual 1 fs chemistry gap *browser-native*—revising an earlier conclusion that the gap was an inter-track effect requiring a native runtime—and we document the self-caught analysis error that briefly obscured this, as a demonstration of the falsifiable-ledger discipline.

Data and code availability

Source, the full validation ladder, and every JSON artifact are at <https://github.com/abgnydn/webgpu-dna>; a live build runs at <https://webgpudna.com>. A versioned Zenodo deposit accompanies release v0.4.0 (DOI to be inserted). Each experiment reruns via `npm run experiments - <id>`.

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